

Solutions to exercises in Lecture 5

Q1) Recall lecture 2 where we wrote down the pseudo-code for the Jacobi iteration. We will follow the same style, but now for FTCS algorithm.

set $n = 0$ where n defines the time iteration

initialise T_j^0 (this initial condition is usually given in the problem)

$$\text{calculate } \Gamma_x = \frac{k \Delta t}{\Delta x^2}$$

repeat

 apply boundary conditions

 update using $T_j^{n+1} = T_j^n + \Gamma_x (T_{j+1}^n - 2T_j^n + T_{j-1}^n)$ for all j

 increase n by 1.

until a "stop" criterion is met, e.g. a certain n value

Q2) If a central difference scheme is used for the time derivative, we have

$$\frac{T_j^{n+1} - T_j^{n-1}}{2\Delta t} = \frac{k}{\Delta x^2} (T_{j+1}^n - 2T_j^n + T_{j-1}^n)$$

$$T_j^{n+1} = T_j^{n-1} + \frac{2k\Delta t}{\Delta x^2} (T_{j+1}^n - 2T_j^n + T_{j-1}^n)$$

writing $\Gamma_x = \frac{2k\Delta t}{\Delta x^2}$ and using the definition for the error

$$\xi(x, t) = \sum E_m(t) e^{i\lambda_m x}$$

as in the lecture we obtain

$$E_m(t+\Delta t) e^{i\lambda_m x} = E_m(t-\Delta t) e^{i\lambda_m x} + \Gamma_x E_m(t) e^{i\lambda_m x} [e^{i\lambda_m \Delta x} + e^{-i\lambda_m \Delta x} - 2]$$

Dividing by $E_m(t) e^{i\lambda_m x}$, we can rewrite the above as

$$\frac{E_m(t+\Delta t)}{E_m(t)} = \frac{E_m(t-\Delta t)}{E_m(t)} + \Gamma_x [2 \cos \lambda_m x - 2]$$

We now make the assumption that the ratio of the amplitude E_m over corresponding time steps is the same: $g = \frac{E_m(t+\Delta t)}{E_m(t)} = \frac{E_m(t)}{E_m(t-\Delta t)}$.

Thus:

$$g = \frac{1}{g} - 4\Gamma_x \sin^2\left(\frac{k_m \Delta x}{2}\right)$$

$$g^2 + g \left[4\Gamma_x \sin^2\left(\frac{k_m \Delta x}{2}\right) \right] - 1 = 0$$

$$g = \frac{-4\Gamma_x \sin^2\left(\frac{k_m \Delta x}{2}\right) \pm \sqrt{16\Gamma_x^2 \sin^4\left(\frac{k_m \Delta x}{2}\right) + 4}}{2}$$

The scheme is only stable if $|g|$ is always less than 1. We see this is untrue if we take the $(-)$ solution from the above equation and take $\sin^2\left(\frac{k_m \Delta x}{2}\right) = 1$. We get

$$g = \frac{-4\Gamma_x - \sqrt{16\Gamma_x^2 + 4}}{2} = -\sqrt{1 + 4\Gamma_x^2} - 2\Gamma_x$$

$|g| > 1$ as long as $\Gamma_x > 0$.

So the scheme is unconditionally unstable

(Q3) For 2D FTCS scheme, we have Eq (4) in the lecture note.

We can carry out von Neumann stability analysis using

$$\xi(x, y, t) = \sum E_{m,n}(t) e^{ik_m x} e^{ik_n y}$$

Substituting this and following the same steps we get $\Gamma_x = \frac{k_2 \Delta t}{\Delta x^2}$ & $\Gamma_y = \frac{k_2 \Delta t}{\Delta y^2}$

$$\begin{aligned} g &= \frac{E_{m,n}(t+\Delta t)}{E_{m,n}(t)} = 1 + \Gamma_x (e^{ik_m \Delta x} + e^{-ik_m \Delta x} - 2) + \Gamma_y (e^{ik_n \Delta y} + e^{-ik_n \Delta y} - 2) \\ &= 1 - 4\Gamma_x \sin^2\left(\frac{k_m \Delta x}{2}\right) - 4\Gamma_y \sin^2\left(\frac{k_n \Delta y}{2}\right) \end{aligned}$$

For the scheme to be stable $|g| \leq 1$. The most restrictive scenario is when $\sin^2\left(\frac{k_m \Delta x}{2}\right) = 1$ and $\sin^2\left(\frac{k_n \Delta y}{2}\right) = 1$. Hence to be stable we need

$$\begin{aligned} -1 + 4\Gamma_x + 4\Gamma_y &\leq 1 \\ \Gamma_x + \Gamma_y &\leq 1/2 \end{aligned}$$